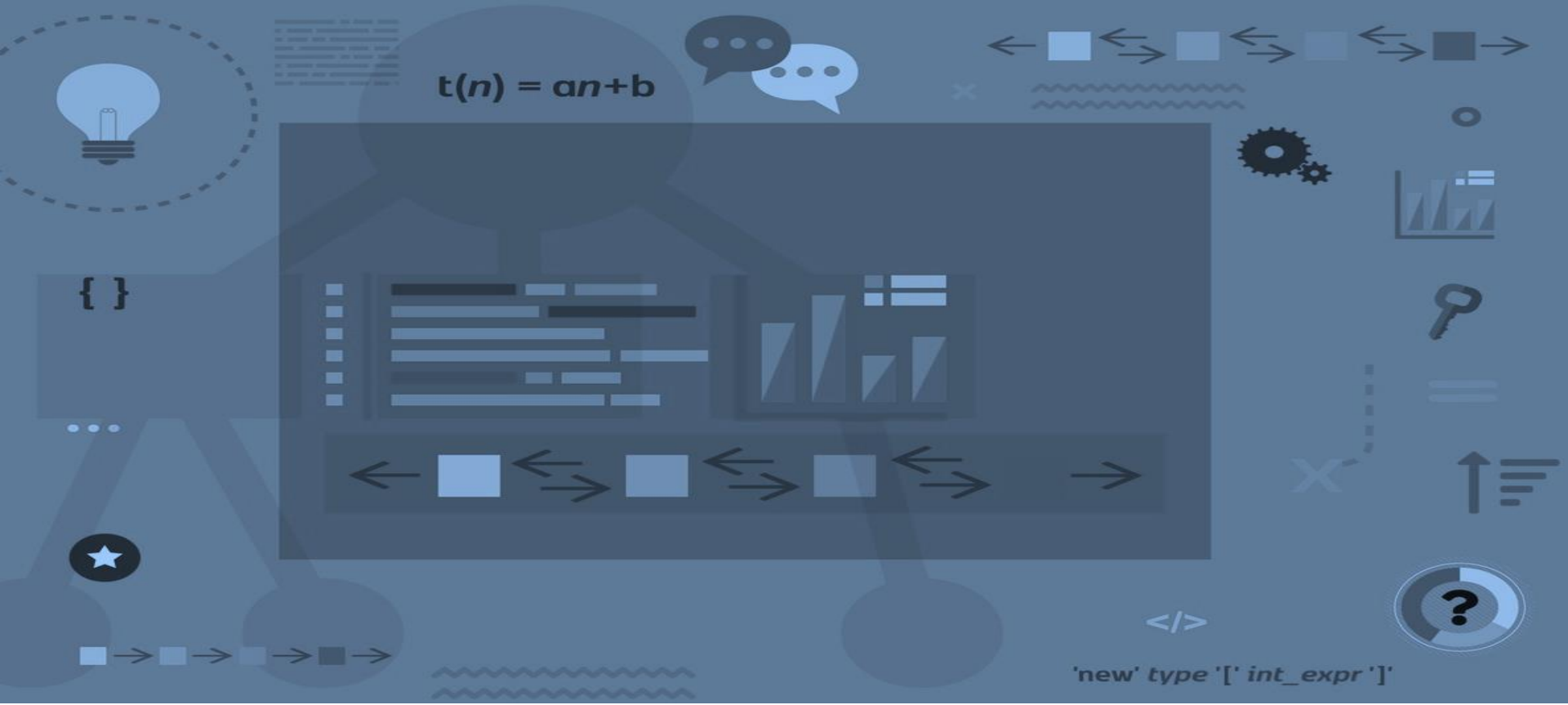

Java Programming

CPT111 – Lab 1

Teng Ma

Exercise 1

Write a Java program that prints "I love you 3000."
Note that that includes the two double quotes.

Exercise 1

```
public class LoveThreeThousands {
    public static void main(String[] args) {
        int threeThousands = 3000;
        System.out.println(direct output?);
    }
}
```

- Key:
  - We could print the sentence directly or we could use a variable
  - How to print double quotes

Exercise 1

```
public class LoveThreeThousands {
    public static void main(String[] args) {
        int threeThousands = 3000;
        System.out.println("\"I love you " + threeThousands + ".\"");
    }
}
```

- Reflection:
  - Use + to merge the output of different parts
  - Use \ to escape characters

Lab exercise 1

Write a Java program that uses basic math to print computation results as in the test case below.

Test case 1:

Output:

num1 + num2 = 13

num1 / num2 = 3

real1 + real2 = 13.0

real1 / real2 = 3.3333333333333335

Lab exercise 1

```
public class SimpleMath {
    public static void main(String[] args) {
        int num1 = 10, num2 = 3;
        double real1 = 10.0 , real2 = 3.0;
        System.out.println(text part + value part);
        So on...
    }
}
```

- Key:
  - How to print the text and value in a single statement
  - How to do the arithmetic in the println() method

Lab exercise 1

```java
public class SimpleMath {
    public static void main(String[] args) {
        int num1 = 10, num2 = 3;
        double real1 = 10.0 , real2 = 3.0;
        System.out.println("num1 + num2 = " + (num1 + num2));
        System.out.println("num1 / num2 = " + (num1 / num2));
        System.out.println("real1 + real2 = " + (real1 + real2));
        System.out.println("real1 / real2 = " + (real1 / real2));
    }
}
```

- Reflections:
  - The use of brackets to change the processing sequence
  - What is the difference between the output of int and double

Lab exercise 2

Write a Java program that prints characters depicting a Christmas tree as in the test case below.
Note that you will need to use escape characters.

Test case 1:
Output:

```
   *
  / \
 /   \
/     \
--| |--
-------
```

Lab exercise 2

```
public class CharXmasTree {
    public static void main(String[] args) {
        System.out.println(blanks + symbol + blanks(optional));
        So on…
    }
}
```

- Key:
  - How to combine blanks and symbols to make them placed correctly
  - How to use escape character to print special characters

Lab exercise 2

```
public class CharXmasTree {
    public static void main(String[] args) {
        System.out.println("   *  ");
        System.out.println("  / \\  ");
        System.out.println(" /   \\  ");
        System.out.println("/     \\  ");
        System.out.println("--| |--");
        System.out.println("-------");
    }
}
```

- Reflections:
  - Can you make it a bigger tree?

Lab CHALLENGE!

- Except for the escape character \, there are several other characters could be used to manipulate the output (have a try to see what they do!):
  - \t
  - \n
  - \r
  - A set of Unicode characters like © (\u00A9), ¥ (\u00A5), etc… And even for Chinese characters
  - Please find the Unicode of Chinese characters here:
    - https://www.chineseconverter.com/en/convert/unicode
    - A bit slow, since there are so many Chinese characters
- Now please use these characters in the System.out.println() statement, and the specific tasks are as shown in the next page

Lab CHALLENGE! - Tasks

- Please write the code to print the output in the following format:

```
My name is
马腾
and I am learning	CPT111
```

- Requirement:
  - There must a single line of statement for output (there is just one line of System.out.println())
  - The space before CPT111 is a tab stop (find which special character does this)
  - And feel free to explore more interesting usage of characters in Java
    - Emojis, characters in other languages

Thank you for your attention !

- Thanks for you effort!
- Feel free to ask questions about lab materials, lectures, and IDE installation.
- You can leave the lab when you complete the lab challenge and ASSESSED by TA already!